

# Project 2: Data Visualization Tool

The **Data Visualization Tool** is an interactive application designed to help users analyze and interpret complex datasets through graphical representations. This tool supports various chart types, including bar graphs, pie charts, and line plots, enabling businesses and researchers to identify trends and make data-driven decisions.

Built using **JavaScript with D3.js** for dynamic visualizations and **Python (Flask)** for backend processing, the tool allows users to upload datasets in **CSV or Excel** formats and generate customized visualizations. The intuitive interface ensures that even non-technical users can easily explore and present data effectively.

The primary objective of this project is to simplify data analysis by transforming raw data into meaningful insights. By using this tool, organizations can enhance reporting, improve decision-making, and communicate findings more effectively through clear and engaging visuals.

